

CS 736

Medical Image Computing (MIC)

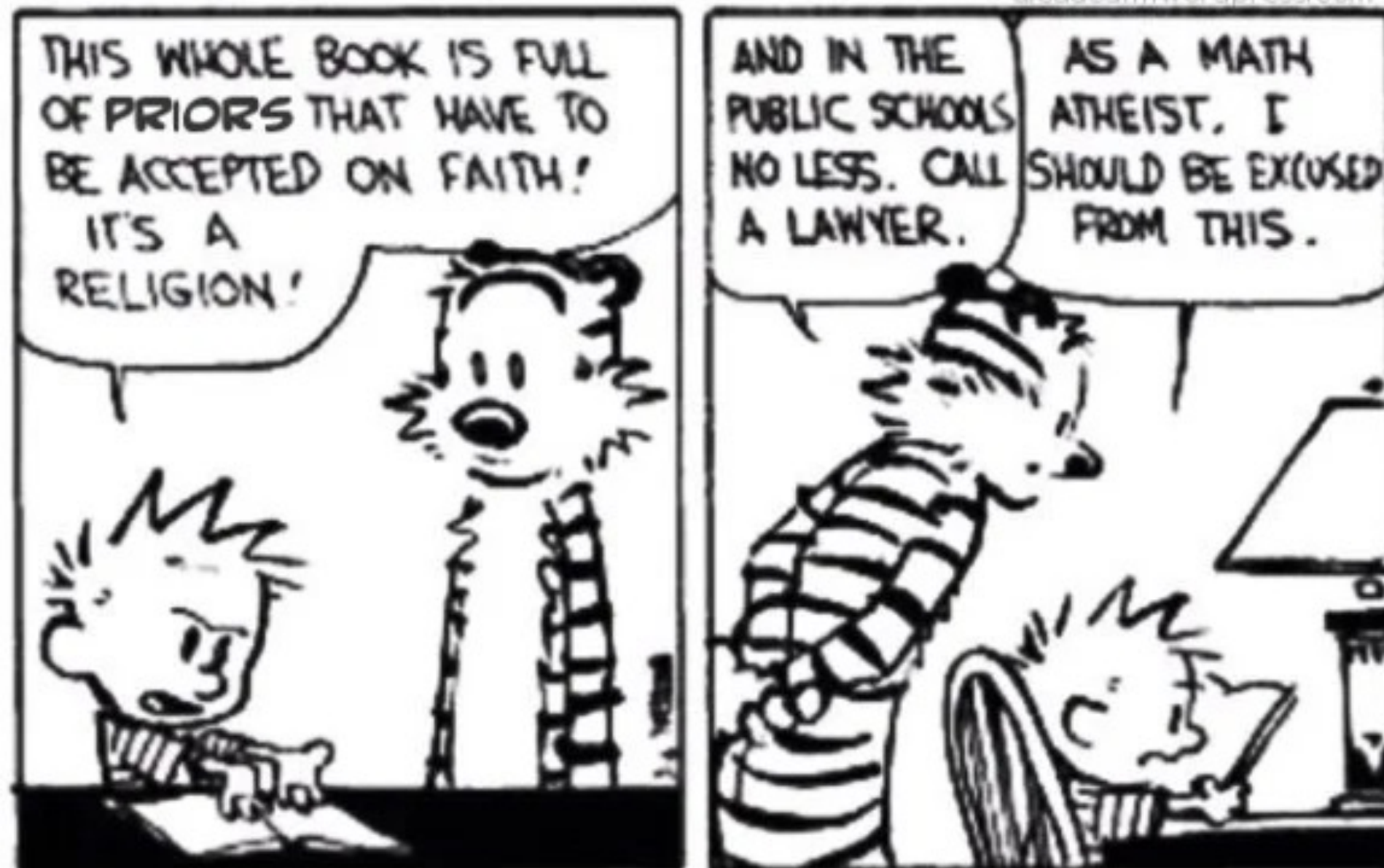
Image Denoising with MRF Priors

Suyash P. Awate

Image Denoising

BAYESIAN INFERENCE

aleadeum.wordpress.com



A P R Advances in Pattern Recognition



**Markov
Random Field
Modeling in
Image Analysis**

Stan Z. Li

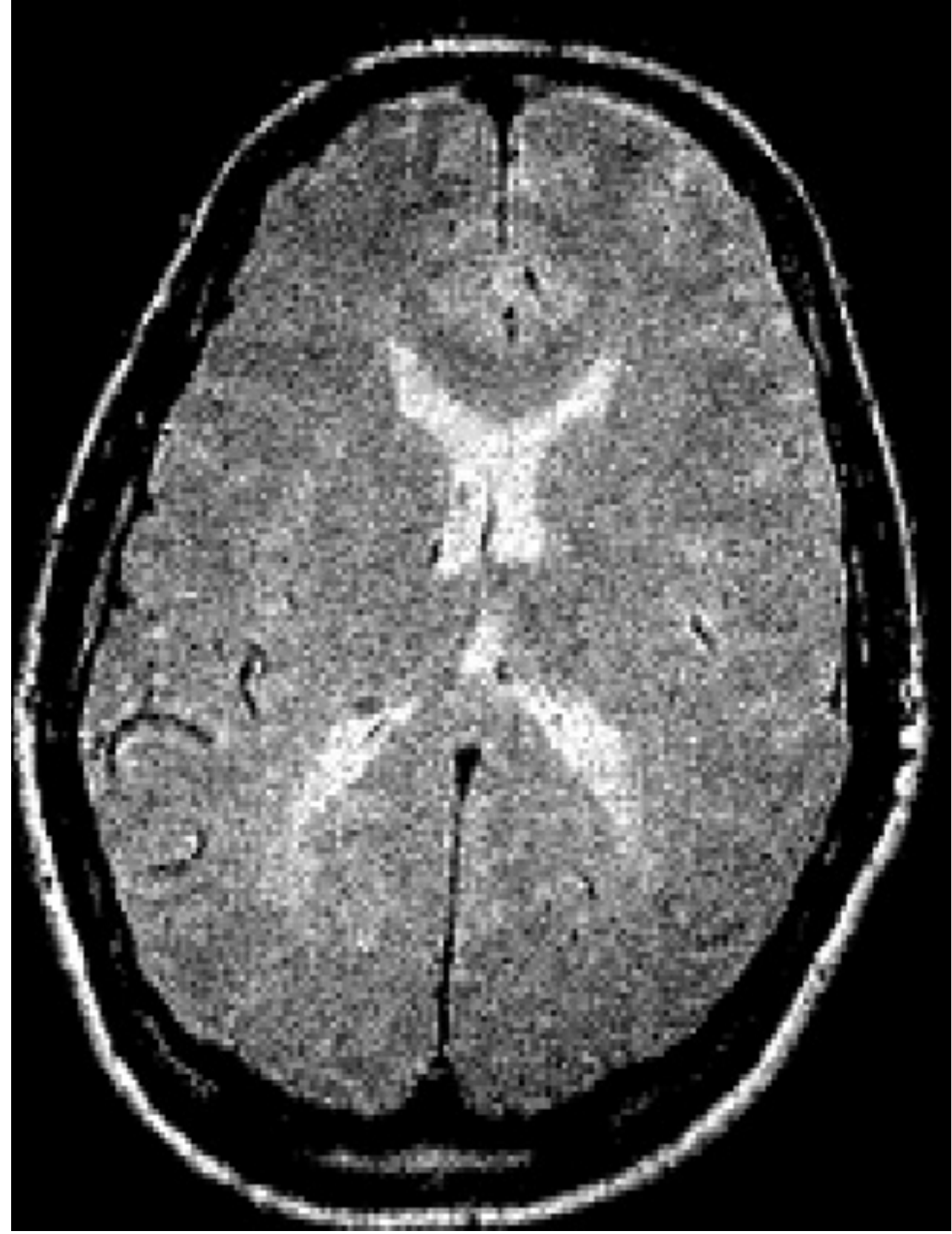


Third Edition

 Springer

Image Denoising

- Noise in image of brain, obtained via magnetic resonance imaging (MRI)



C: 35.0, W: 350

-

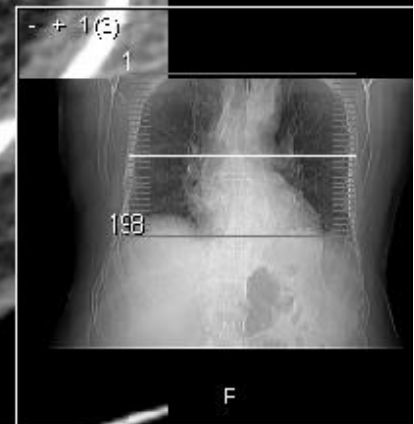


Image Denoisi

- Noise in image of fetus, obtained using ultrasound



Image Denoising

- Simulated Gaussian noise in a 2D image
 - Independent and identically-distributed (i.i.d.) across pixels

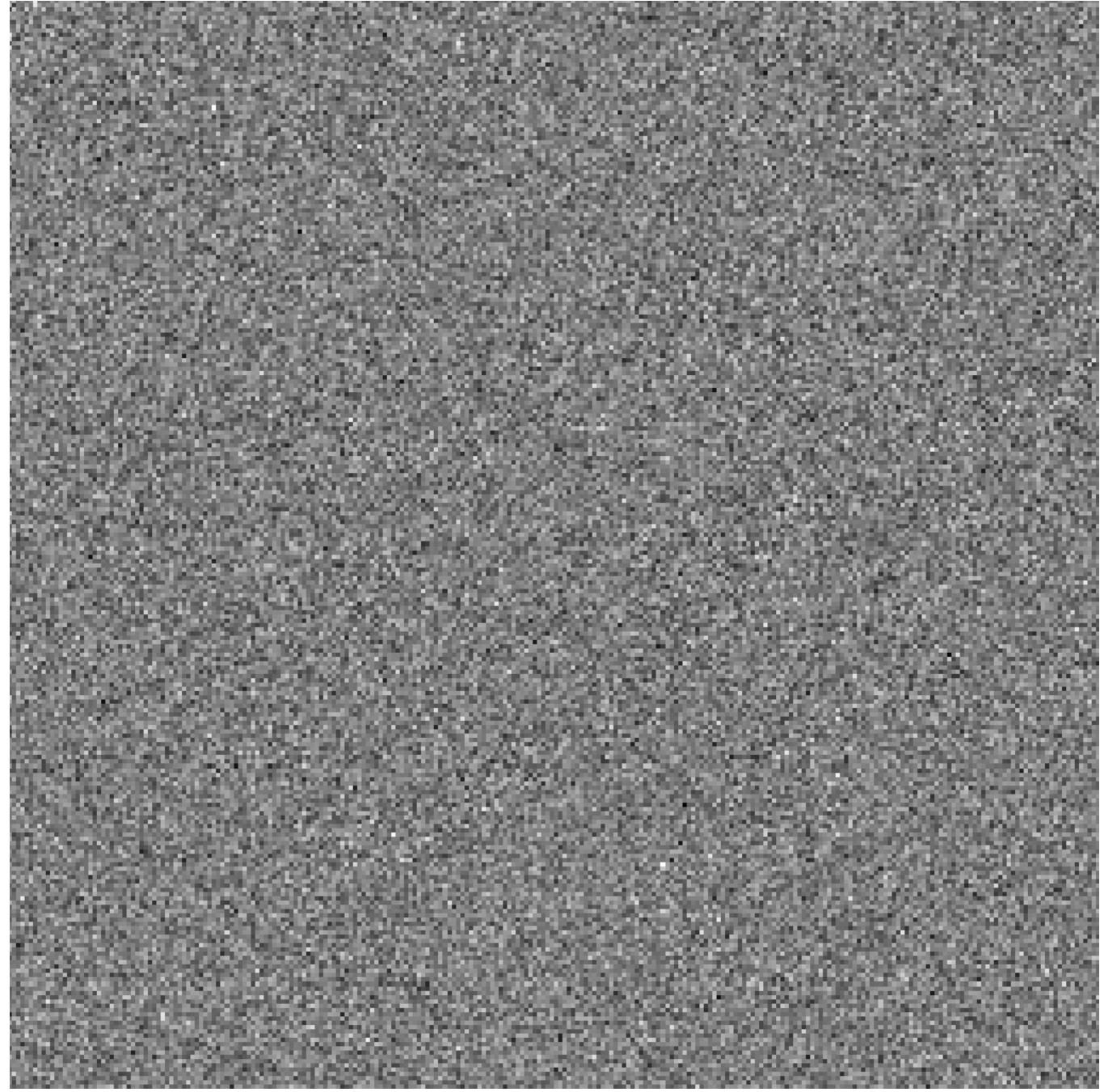
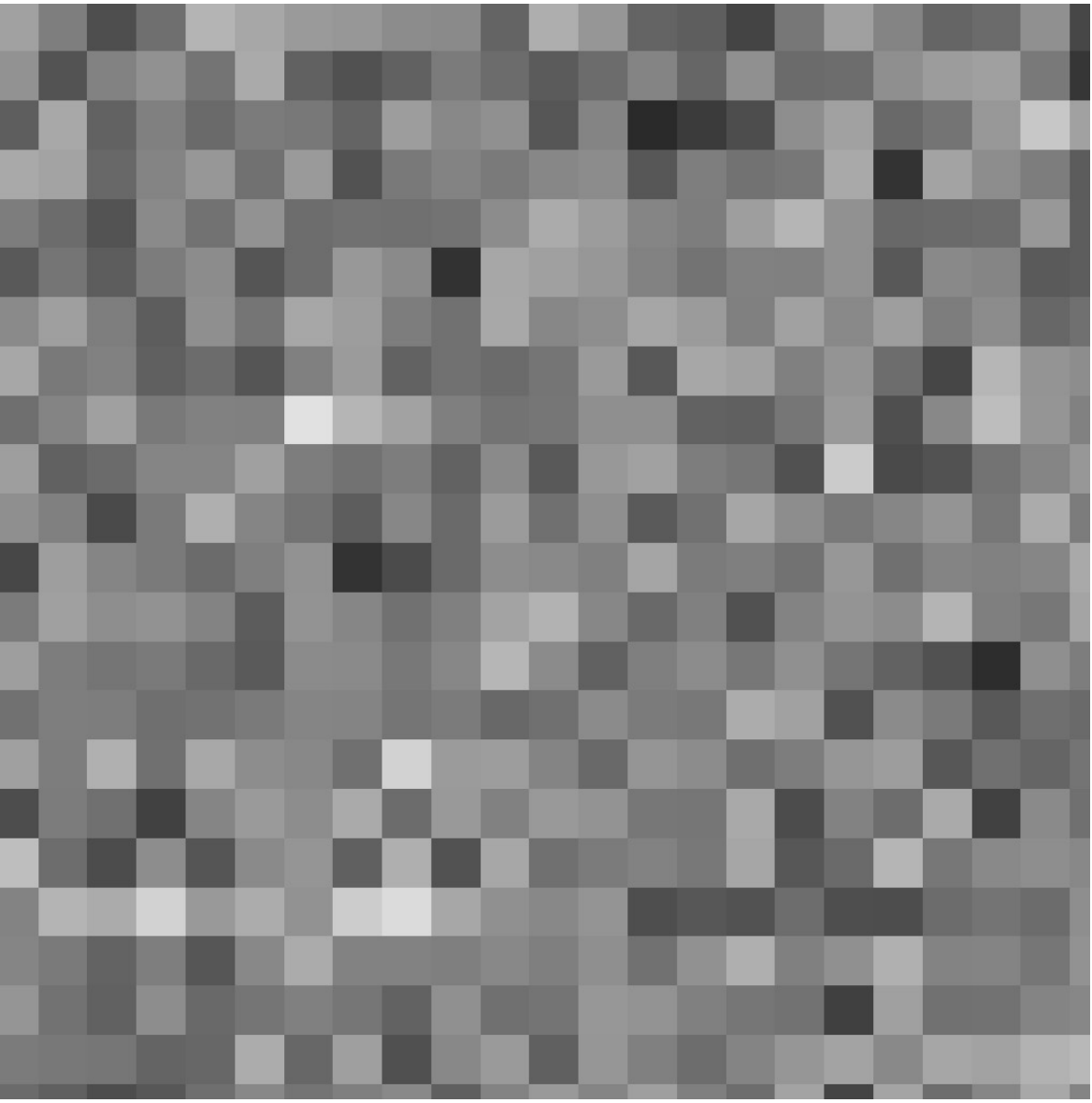


Image Denoising

- Simulated correlated-Gaussian noise in a 2D image
 - Take an image of i.i.d. Gaussian noise
 - Then,
apply smoothing (low-pass) filter,
e.g., mean filter
or Gaussian convolution

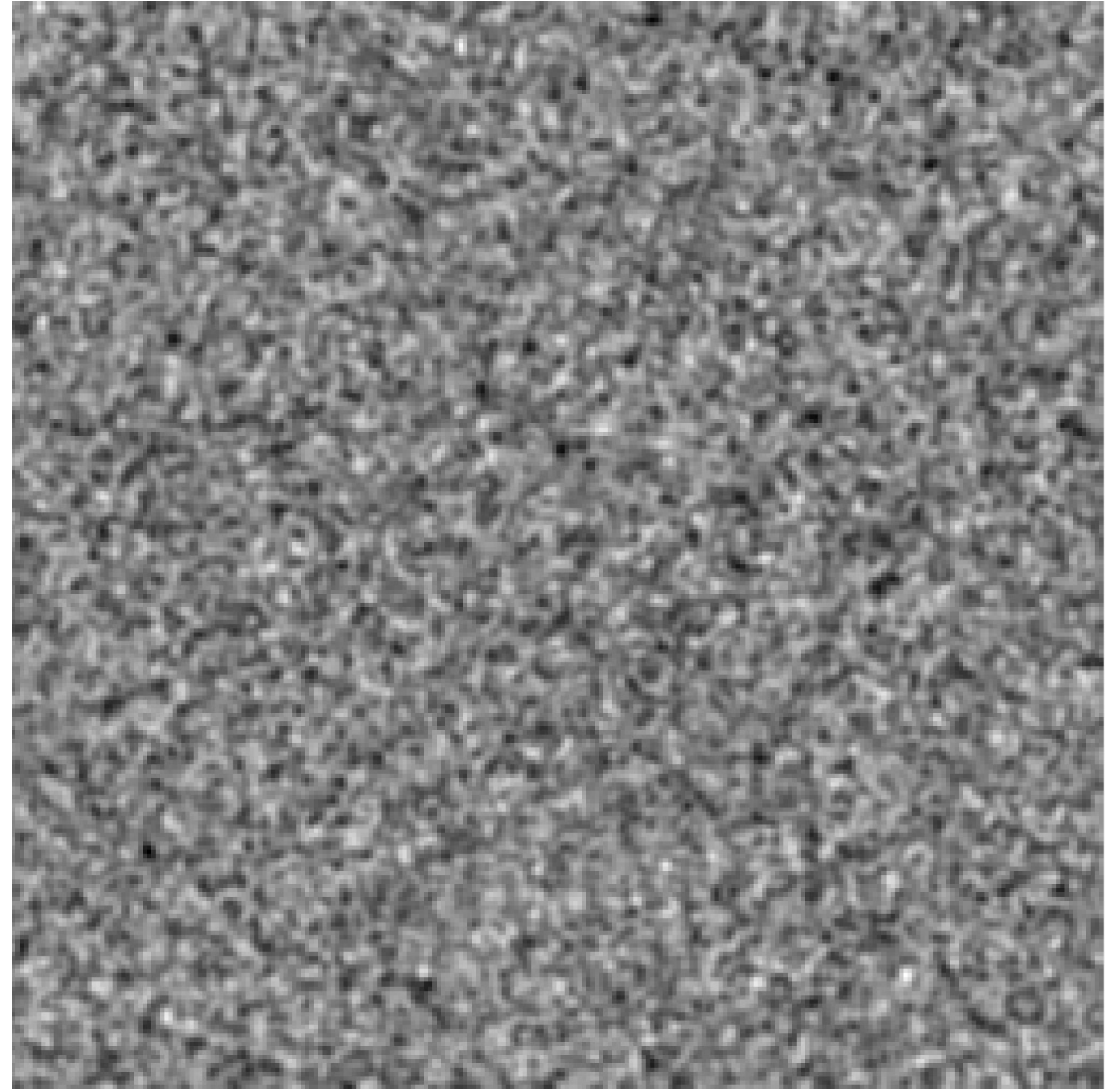


Image Denoising

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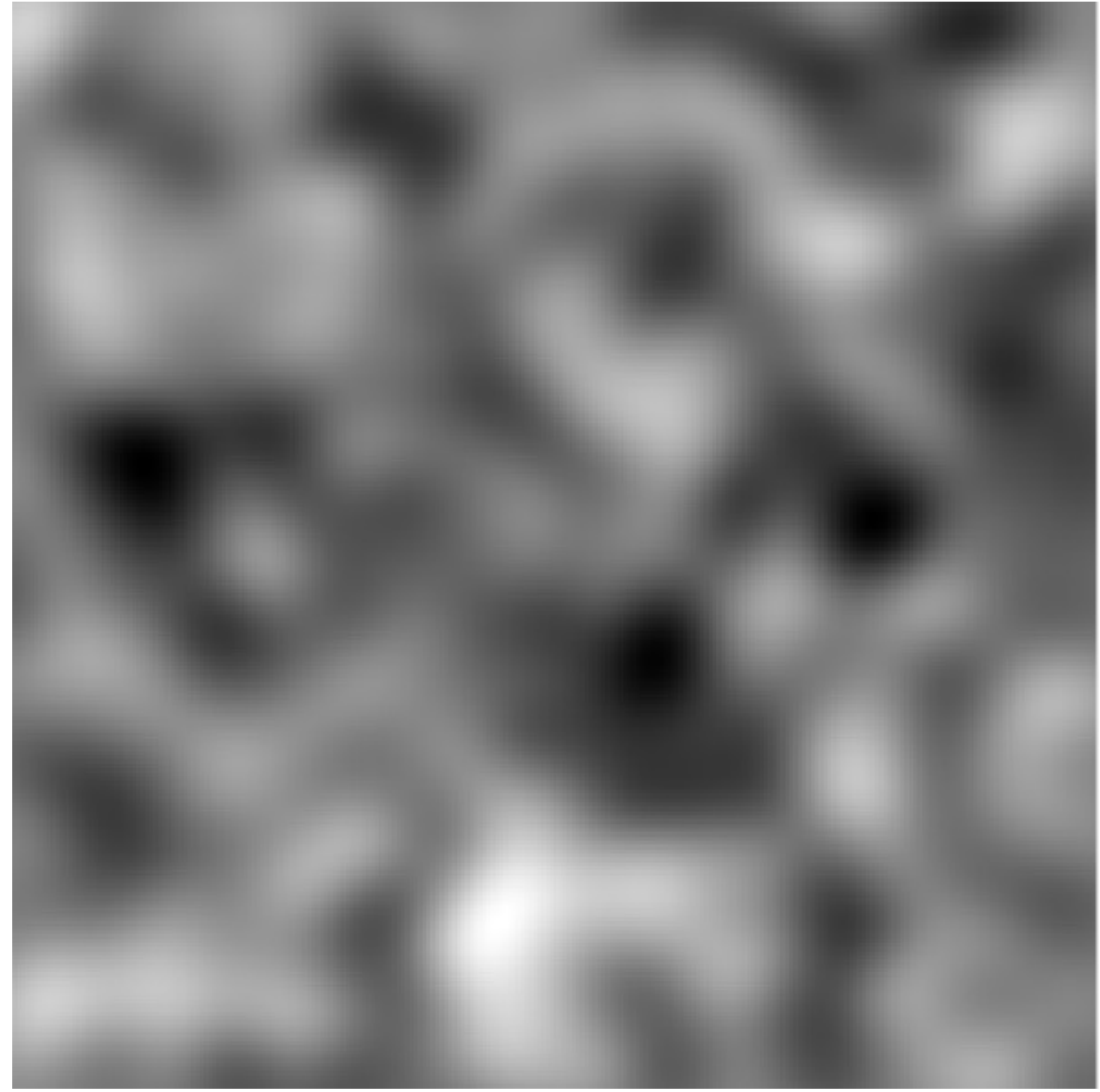


Image Denoising

- Noise models in medical imaging
 - Let random-variable X model observed intensity at a pixel/voxel
 - For random variable X , **signal-to-noise ratio (SNR)** equals $E[X]/SD(X)$

Image Denoising

- Observing data



Image Denoising

- Noise models in medical imaging
 - Most common is **Gaussian** because of:
 - 1) Central limit theorems for:
 - Means of i.i.d random variables
 - But also for some cases of sums of not-independent or unidentically-distributed variables [1,2,3] or both
 - 2) Thermal noise in electronic systems
 - Thermal agitation of charge carriers
 - Inevitable at non-zero temperature
 - 3) Brownian motion, random walks
 - SNR for a Gaussian random variable
 - Mean-parameter / standard-deviation-parameter

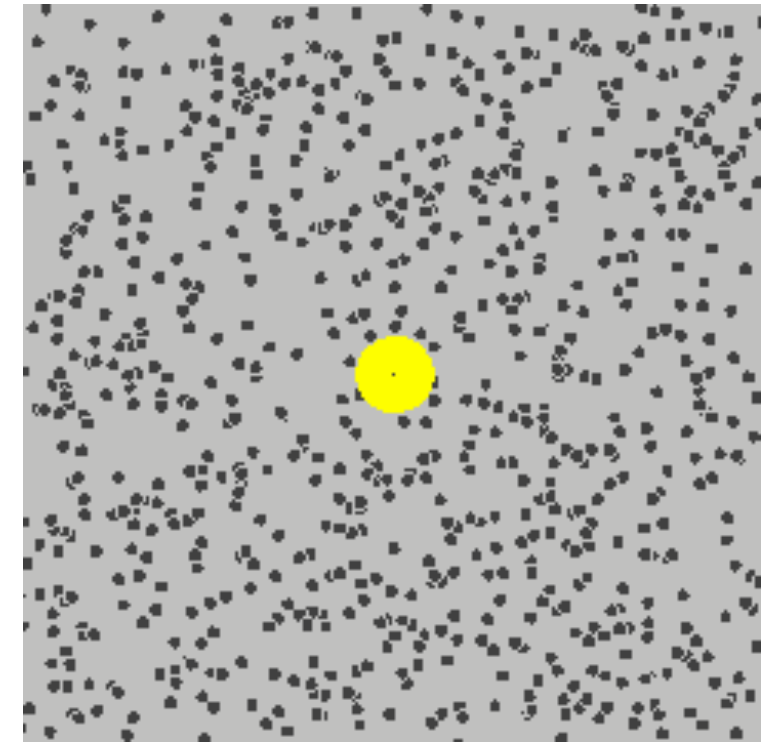
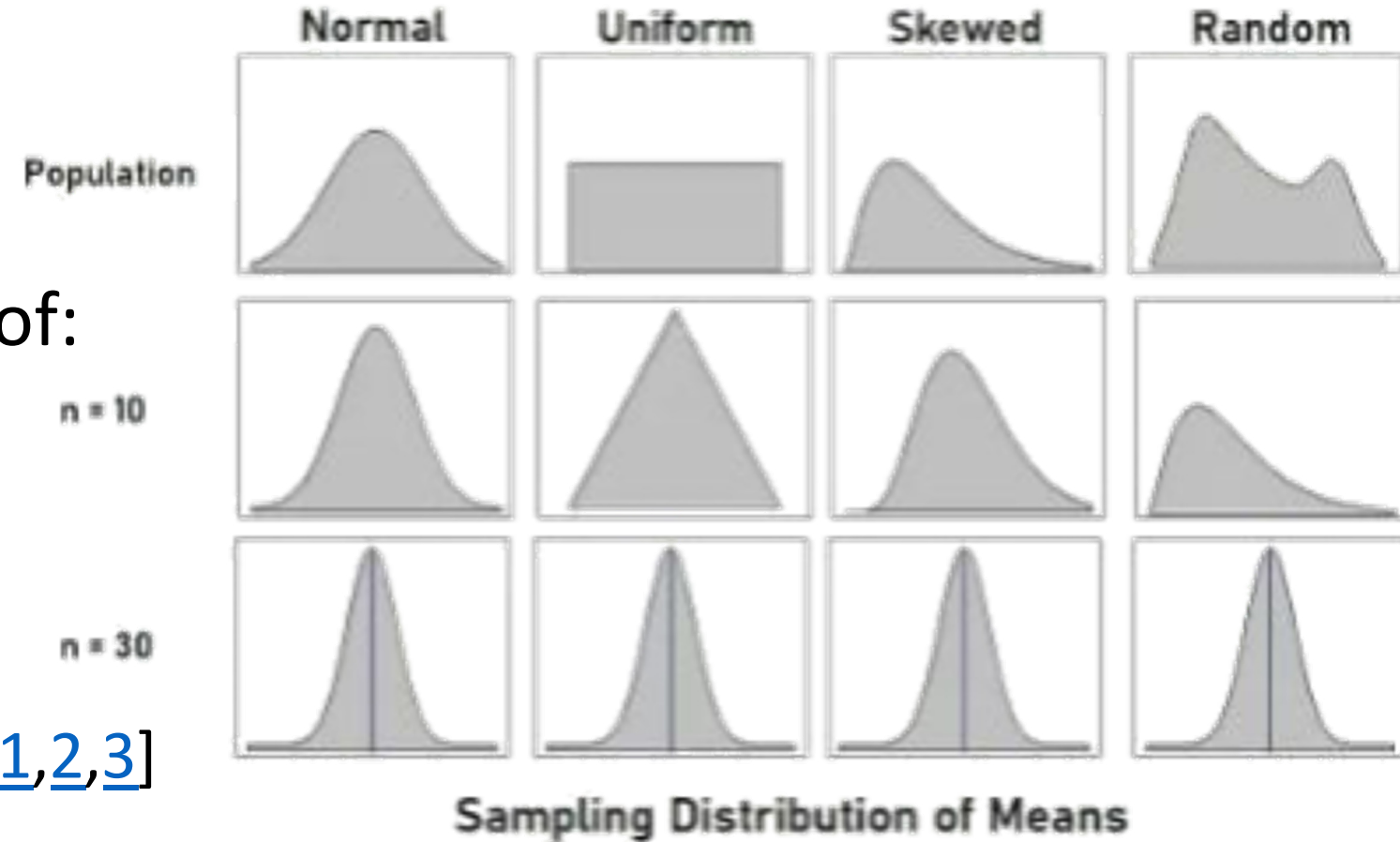


Image Denoising

- Johann Carl Friedrich Gauss (1777–1855)
 - “Foremost of mathematicians”,
“Greatest mathematician since antiquity”
 - Number theory, algebra, statistics, analysis, differential geometry, geophysics, electrostatics, astronomy, optics
 - Child prodigy
 - Published Arithmetical Investigations (age 21), consolidating number theory as a discipline
 - PhD in 1799 (age 22)
 - Gave a new proof of fundamental theorem of algebra
 - Every non-zero, single-variable, **degree- n polynomial** with complex coefficients has, counted with multiplicity, exactly **n roots**
 - FFT was invented by Gauss in 1805 (age 28)
 - Cooley and Tukey reinvented it in 1965 and popularized it



Image Denoising

- Johann Carl Friedrich Gauss (1777–1855)
 - Refused to publish “unripe” work (“few, but ripe”)
 - Personal diary indicates discoveries before contemporaries
 - Declined to provide intuition behind his elegant proofs
 - Erased all traces of discovery process
 - Wanted proofs to look 'magical'
 - All analysis must be suppressed for sake of brevity
 - Students
 - Dirichlet, Kirchoff, Riemann, Mobius
 - www.youtube.com/watch?v=q6N2FeXoMvg



Image Denoising

- Noise models in medical imaging: **Poisson**
- When measurement made by accumulation of photons over a detector
 - e.g., photography, light microscope, infrared imaging, X-ray, CT
- Given: Average number of captured photons (parameter) $\lambda \in \mathbb{R}$ during a certain time interval of imaging
 - λ can be non-integer
- Probability of 'k' captured photons:
$$P(X = k; \lambda) = \frac{\lambda^k \exp(-\lambda)}{k!}$$
- SNR for a Poisson random variable
 - $E[X] = \text{Var}[X] = \lambda$
 - $\text{SNR}(X) = E[X] / \text{SD}(X) = \sqrt{\lambda}$

Image Denoising

- Siméon Denis Poisson
- Mathematician, geometer, physicist
- Academic advisors
 - Lagrange, Laplace
- Students
 - Dirichlet
 - Chasles → HA Newton → JW Gibbs
- www.youtube.com/watch?v=J1uDhmWNMvo



Image Denoising

- Noise models in medical imaging: **Poisson**
- For large λ , PMF shape approaches a Gaussian/bell-like curve

$$P(X = k; \lambda) = \frac{\lambda^k \exp(-\lambda)}{k!}$$

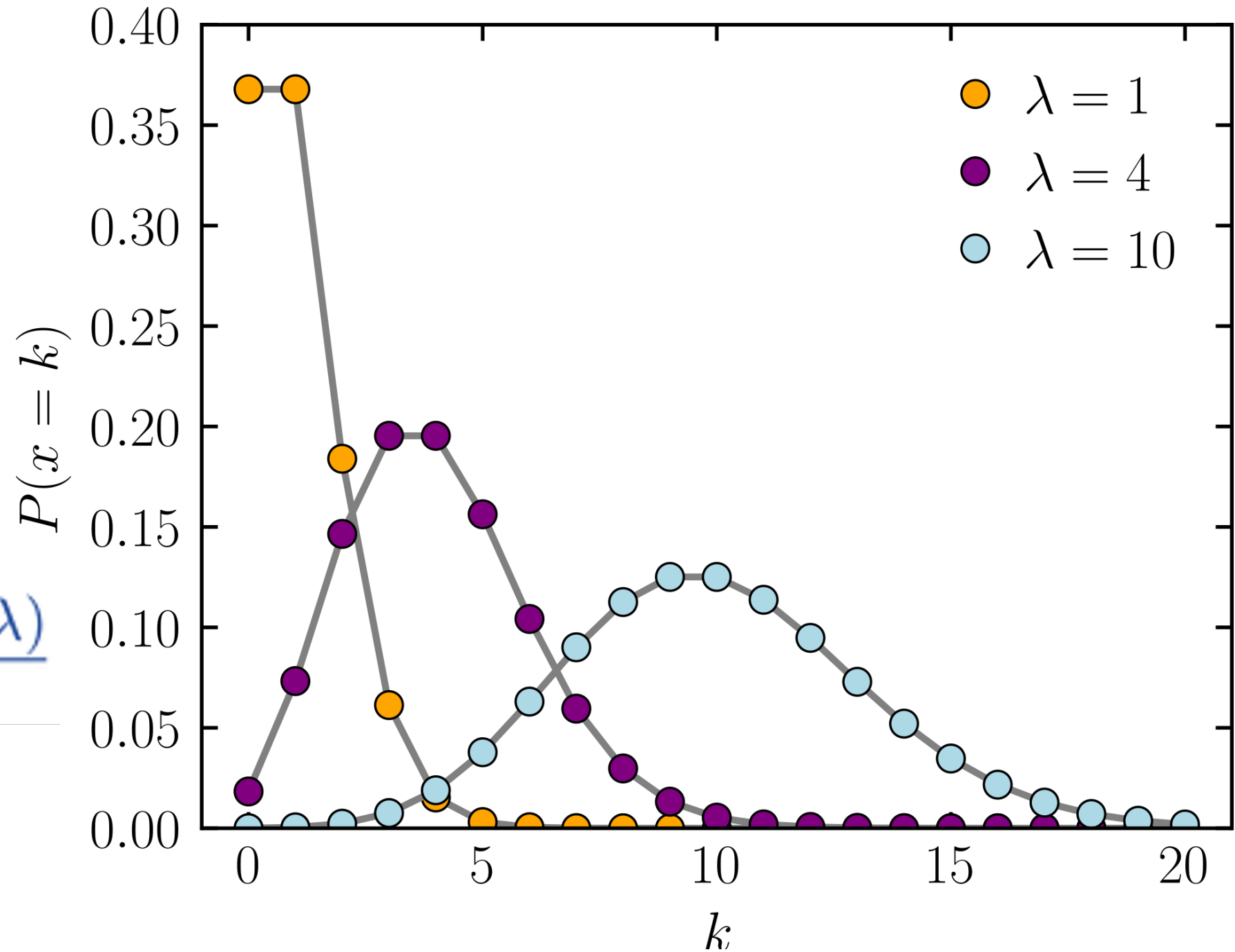


Image Denoising

- Noise models in medical imaging: **Compound Poisson**
- Practical X-ray-based CT uses “polychromatic (broad-beam)” X rays
- Compound-Poisson random variable =
sum of N i.i.d. random variables
where N itself is drawn from a Poisson distribution
 - In case of X-ray CT, one Poisson PMF for each Xray frequency
- May be approximated by a Gaussian functional form [[1](#)]

Image Denoising

- Noise models in medical imaging: **Rician (Rice)**
- MRI datum, at each voxel, is a 2-vector, say, [a,b]
- Noise in each component is i.i.d. zero-mean additive Gaussian
- Distribution over observed/measured vector magnitude 'x' is Rician

$$p(x|\nu, \sigma) = \frac{x}{\sigma^2} \exp\left(-\frac{x^2 + \nu^2}{2\sigma^2}\right) I_0\left(\frac{x\nu}{\sigma^2}\right)$$

- ν (non-negative): magnitude of 2-vector signal (uncorrupted)
 - Not $E[X]$
- σ (non-negative): std. dev. of Gaussian in each component
 - Not Std.Dev.(X)
- $I_0(.)$ = modified Bessel function of 1st kind with order 0
- PDF defined only for non-negative x

Image Denoising

- Noise models in medical imaging: Rician (Rice)
- PDF mass shifted to right of v

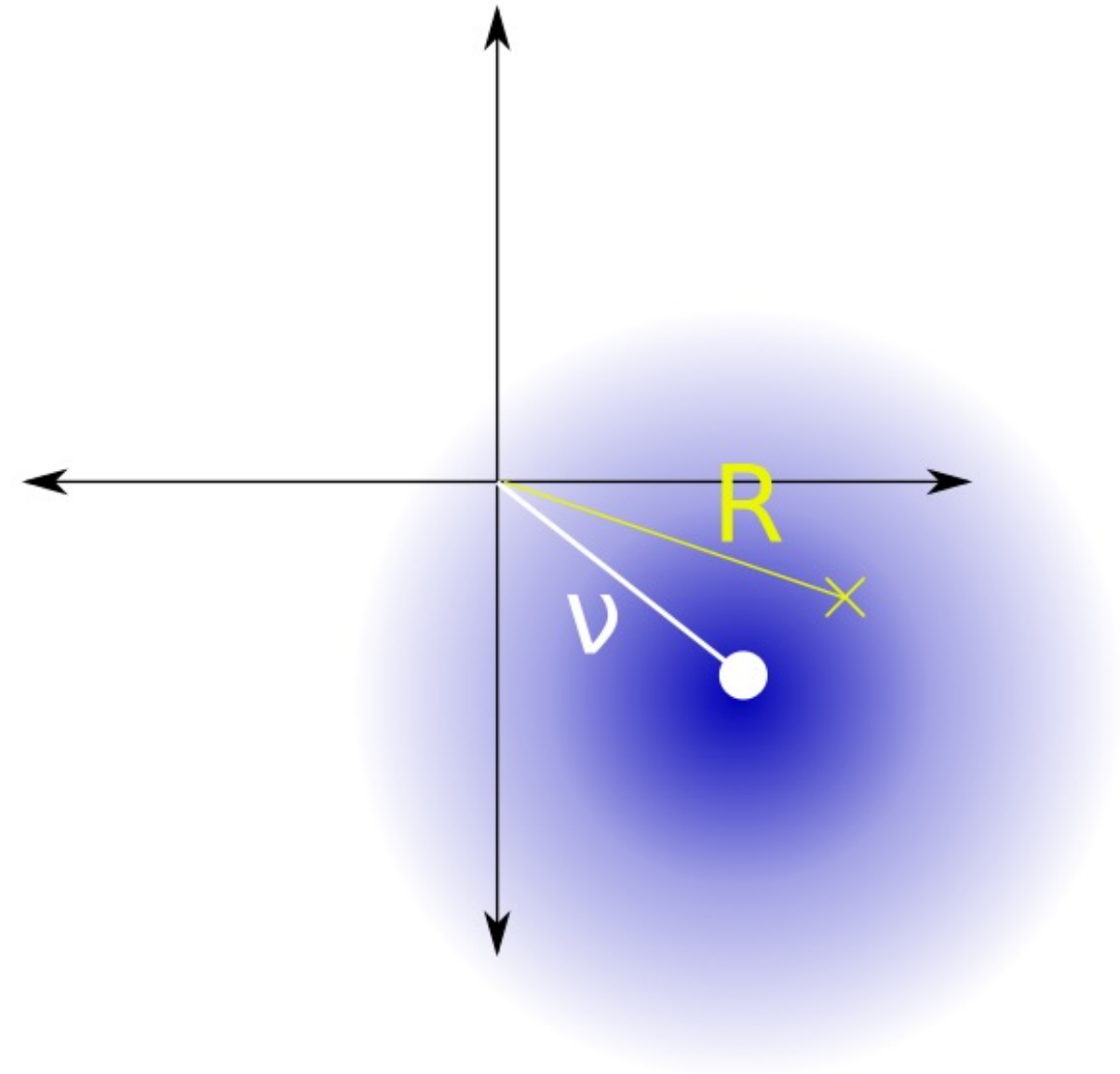
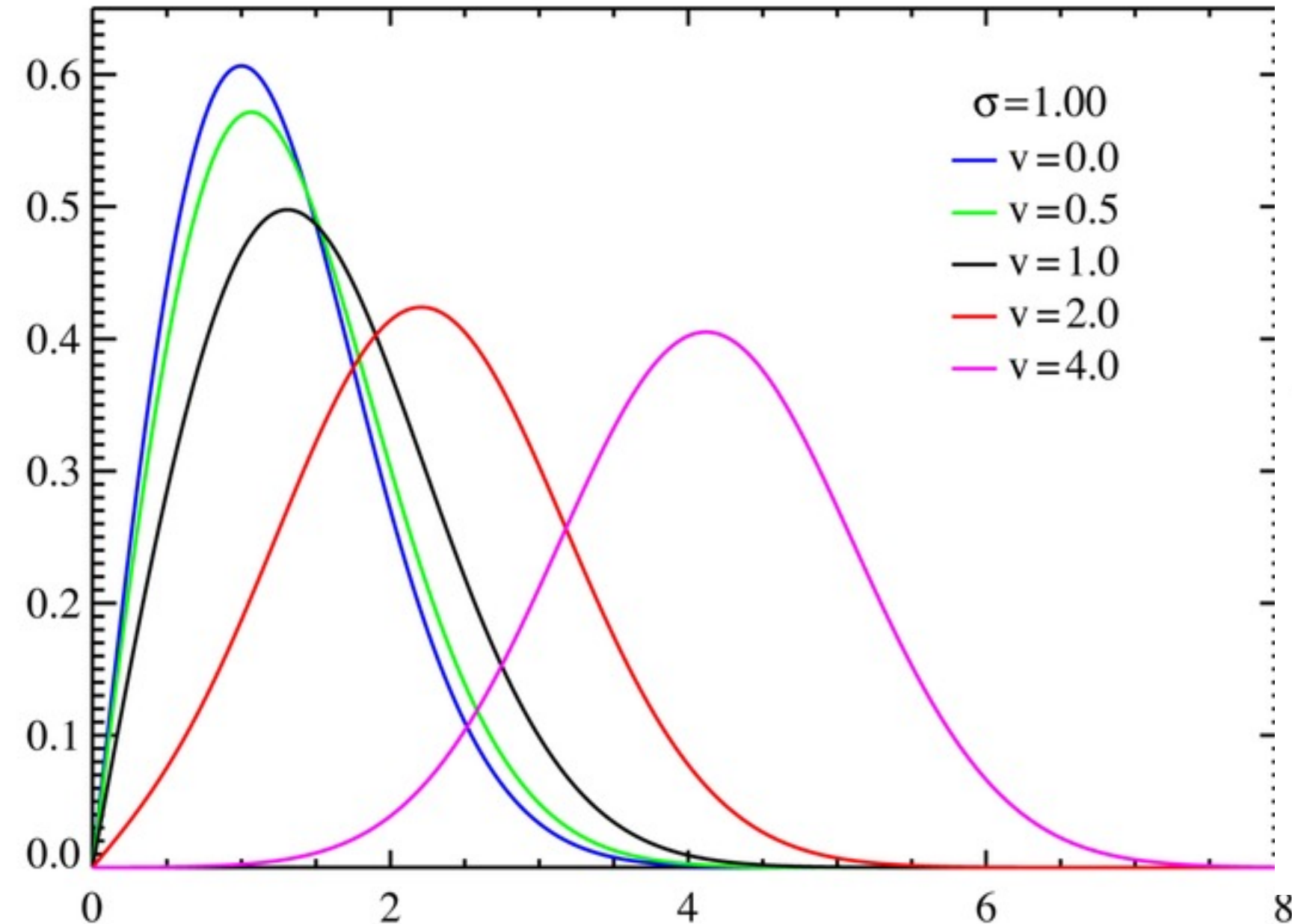
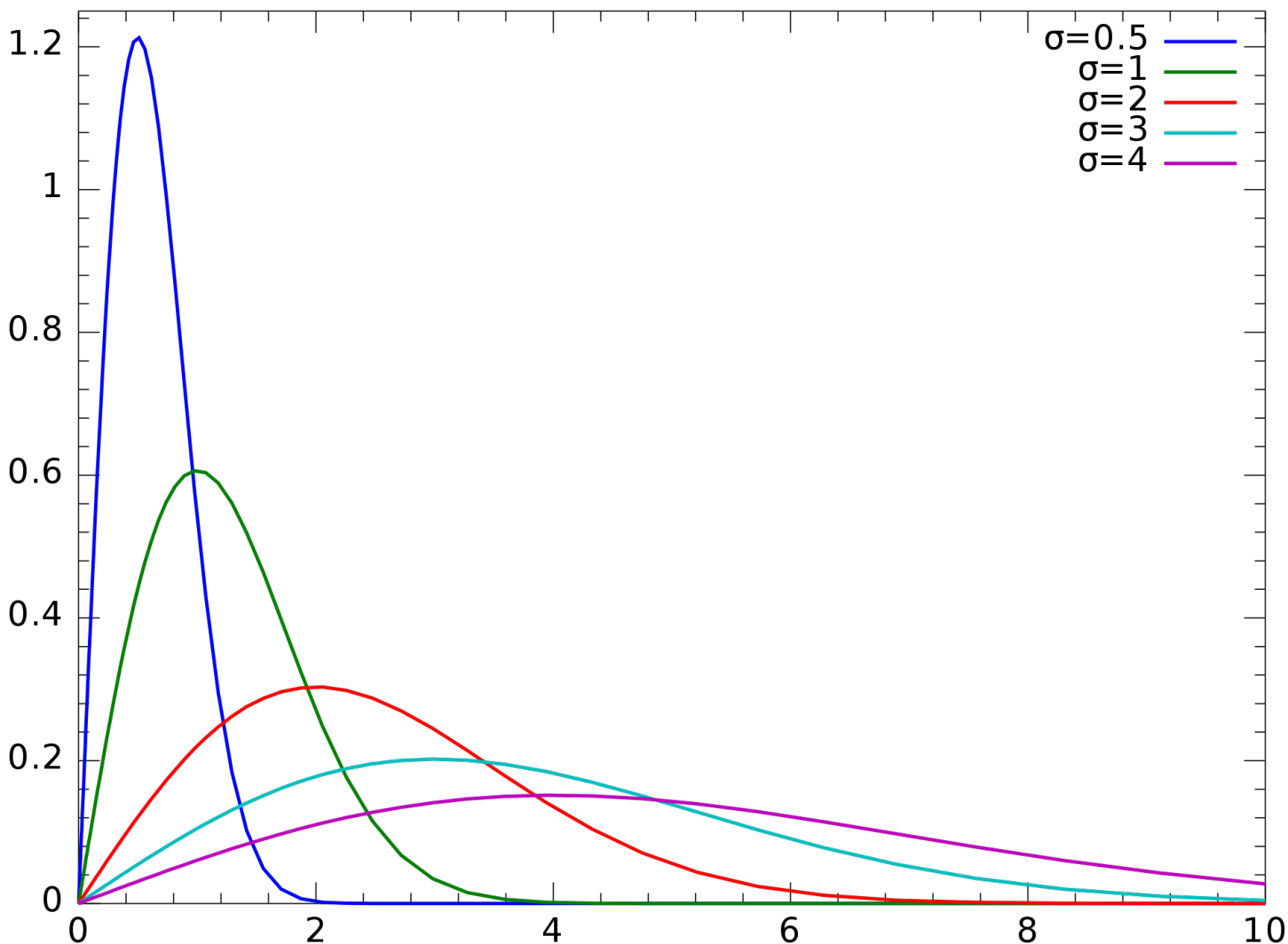


Image Denoising

- Noise models in medical imaging: **Rician (Rice)**

- **Special case: $\nu = 0$**

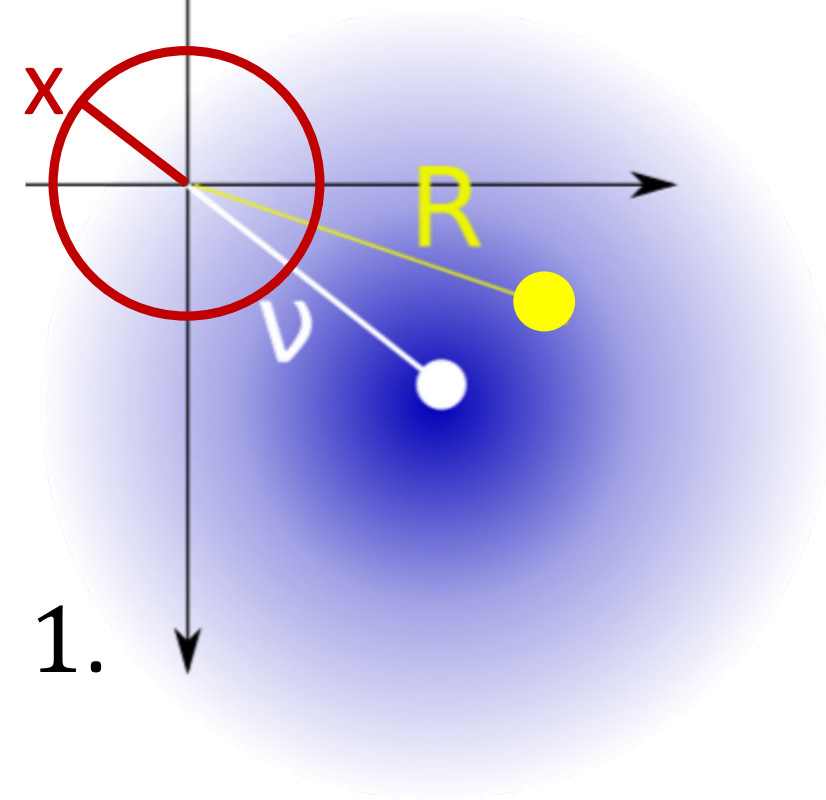
- Rician PDF reduces to the **Rayleigh** PDF: $p(x; \sigma) = \frac{x}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right)$



$$p(x|\nu, \sigma) = \frac{x}{\sigma^2} \exp\left(-\frac{x^2 + \nu^2}{2\sigma^2}\right) I_0\left(\frac{x\nu}{\sigma^2}\right)$$

Image Denoising

- Noise models in medical imaging: **Rician (Rice)**
- True signal (a,b) has magnitude $\nu := \sqrt{a^2 + b^2}$
- Observed signal is $(z_1 \sim N(a, \sigma^2), z_2 \sim N(b, \sigma^2))$ has magnitude $R := \sqrt{z_1^2 + z_2^2}$. For simplicity, $\sigma = 1$.
- Desired CDF $P(R \leq x) = P(z_1^2 + z_2^2 \leq x^2)$



$$= \frac{1}{2\pi} \int_{(z_1, z_2): z_1^2 + z_2^2 \leq x^2} \exp\left(-\frac{z_1^2 + z_2^2}{2} - \frac{a^2 + b^2}{2} + \frac{2z_1 a + 2z_2 b}{2}\right) dz_1 dz_2$$

$$= \frac{1}{2\pi} \int_{\theta=0}^{2\pi} \int_{R=0}^x \exp\left(-\frac{R^2}{2} - \frac{\nu^2}{2} + aR \cos \theta + bR \sin \theta\right) R dR d\theta$$

Apply **Leibniz integral rule** to get PDF $P(R = x) := \frac{\partial}{\partial x} P(R \leq x) =$

$$= \frac{x}{2\pi} \exp\left(-\frac{x^2}{2} - \frac{\nu^2}{2}\right) \int_{\theta=0}^{2\pi} \frac{\partial x}{\partial x} \exp(ax \cos \theta + bx \sin \theta) d\theta$$

Image Denoising

- Noise models in medical imaging: **Rician (Rice)**

$$\begin{aligned} & \bullet \frac{1}{2\pi} \int_{\theta=0}^{2\pi} \exp(a\mathbf{x} \cos \theta + b\mathbf{x} \sin \theta) d\theta \\ &= \frac{1}{2\pi} \int_{\theta=0}^{2\pi} \exp(v \cos \phi \mathbf{x} \cos \theta + v \sin \phi \mathbf{x} \sin \theta) d\theta \\ &= \frac{1}{2\pi} \int_{\theta=0}^{2\pi} \exp(v \mathbf{x} \cos(\theta - \phi)) d\theta \\ &= \frac{1}{2\pi} \int_{\theta'=-\phi}^{2\pi-\phi} \exp(v \mathbf{x} \cos(\theta')) d\theta', \text{ where we substituted } \theta' := \theta - \phi \\ &= \frac{1}{2\pi} \int_{\theta'=0}^{2\pi} \exp(v \mathbf{x} \cos(\theta')) d\theta' \\ &= I_0(v\mathbf{x}), \end{aligned}$$

by definition of modified Bessel function of 1st kind with order 0

Image Denoising

- Noise models in medical imaging:
- **Ultrasound**
- Complicated to characterize because depends on:
 - Device type, e.g., coherent or incoherent waves
 - Scattering density of tissue
- Variety of distributions have been used for magnitude images
 - Rician, Rayleigh, K

Image Denoising

- SNR increases with addition of i.i.d. RVs
- Let X_1, X_2 (independent) have same distribution as X
- $E[X_1 + X_2] = 2 E[X]$
- $\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) = 2 \text{Var}(X)$
- $\text{SD}(X_1 + X_2) = \sqrt{2} \text{SD}(X)$
- $\text{SNR}(X_1 + X_2) = E[X_1 + X_2] / \text{SD}(X_1 + X_2) = \sqrt{2} \text{SNR}(X)$
 - Signal mean increased
 - Signal variance also increased
 - But, increase in mean (relatively) dominates over increase in standard deviation
 - So, SNR increased

Image Denoising

- Bayesian image denoising
- Optimal noiseless image is one that maximizes posterior distribution

$$P(\text{NoiselessImage}|\text{NoisyImage}) = \frac{P(\text{NoisyImage}|\text{NoiselessImage})P(\text{NoiselessImage})}{P(\text{NoisyImage})}$$

- **Likelihood / noise model**
 - Probability of generating data given noiseless image
- **Prior**
 - Our prior beliefs about noiseless image **before** observing data
- **Posterior**: product of likelihood and prior, with normalization
 - What we get “post”/after “observing”/acquiring data

Spatial Interaction and the Statistical Analysis of Lattice Systems

By JULIAN BESAG

University of Liverpool

[Read before the ROYAL STATISTICAL SOCIETY at a meeting organized by the RESEARCH SECTION on Wednesday, March 13th, 1974, Professor J. DURBIN in the Chair]

SUMMARY

The formulation of conditional probability models for finite systems of spatially interacting random variables is examined. A simple alternative proof of the Hammersley–Clifford theorem is presented and the theorem is then used to construct specific spatial schemes on and off the lattice. Particular emphasis is placed upon practical applications of the models in plant ecology when the variates are binary or Gaussian. Some aspects of infinite lattice Gaussian processes are discussed. Methods of statistical analysis for lattice schemes are proposed, including a very flexible coding technique. The methods are illustrated by two numerical examples. It is maintained throughout that the conditional probability approach to the specification and analysis of spatial interaction is more attractive than the alternative joint probability approach.

Keywords: MARKOV FIELDS; SPATIAL INTERACTION; AUTO-MODELS; NEAREST-NEIGHBOUR SCHEMES; STATISTICAL ANALYSIS OF LATTICE SCHEMES; CODING TECHNIQUES; SIMULTANEOUS BILATERAL AUTOREGRESSIONS; CONDITIONAL PROBABILITY MODELS

On the Statistical Analysis of Dirty Pictures

By JULIAN BESAG

University of Durham, U.K.

[Read before the Royal Statistical Society, at a meeting organized by the Research Section on Wednesday, May 7th, 1986, Professor A. F. M. Smith in the Chair]

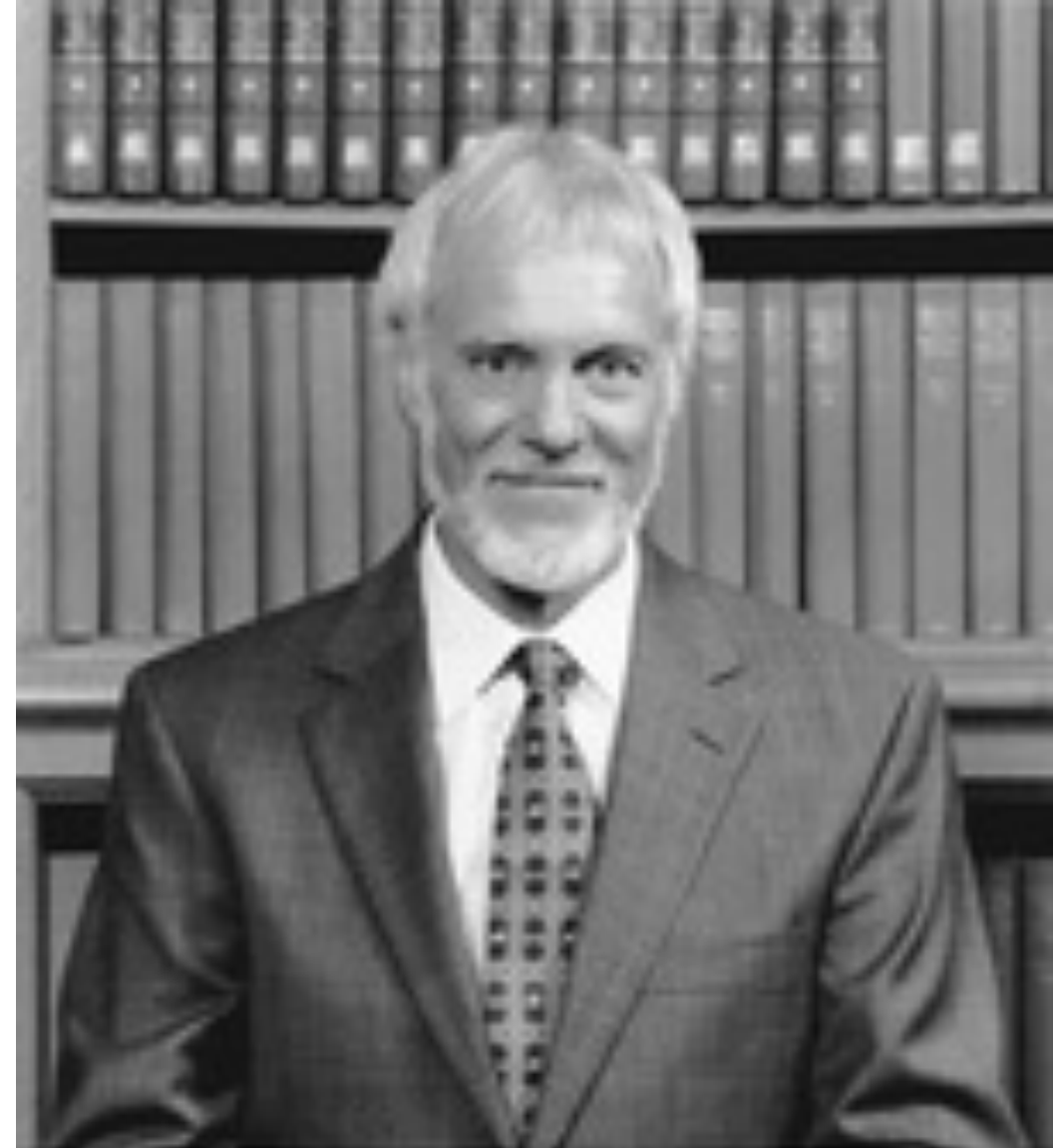
SUMMARY

A continuous two-dimensional region is partitioned into a fine rectangular array of sites or “pixels”, each pixel having a particular “colour” belonging to a prescribed finite set. The true colouring of the region is unknown but, associated with each pixel, there is a possibly multivariate record which conveys imperfect information about its colour according to a known statistical model. The aim is to reconstruct the true scene, with the additional knowledge that pixels close together tend to have the same or similar colours. In this paper, it is assumed that the *local* characteristics of the true scene can be represented by a non-degenerate Markov random field. Such information can be combined with the records by Bayes’ theorem and the true scene can be estimated according to standard criteria. However, the computational burden is enormous and the reconstruction may reflect undesirable large-scale properties of the random field. Thus, a simple, iterative method of reconstruction is proposed, which does not depend on these large-scale characteristics. The method is illustrated by computer simulations in which the original scene is *not* directly related to the assumed random field. Some complications, including parameter estimation, are discussed. Potential applications are mentioned briefly.

Keywords: IMAGE PROCESSING; PATTERN RECOGNITION; SEGMENTATION; CLASSIFICATION; IMAGE RESTORATION; REMOTE SENSING; MARKOV RANDOM FIELDS; PAIRWISE INTERACTIONS; MAXIMUM A POSTERIORI ESTIMATION; SIMULATED ANNEALING; GIBBS SAMPLER; ITERATED CONDITIONAL MODES; GREY-LEVEL SCENES; AUTO-NORMAL MODELS; PSEUDO-LIKELIHOOD ESTIMATION.

1. INTRODUCTION

There has been much recent interest in the following type of problem. A continuous two-dimensional region S is partitioned into a fine rectangular array of sites or picture elements (“pixels”), each having a particular colour, lying in a prescribed set. The colours may be



***J. R. Statist. Soc. B (1986)
48, No. 3, pp. 259–302***

Image Denoising

- **Noiseless** image $X = x$
 - X is a MRF
- **Observed** image (data) $Y = y$
- **Noise model** for observed intensities given noiseless intensities
 - Typically: independent across pixels (doesn't deal with correlated noise)
 - $P(Y | X) := \prod_i P(Y_i | X_i)$
 - e.g., $P(Y_i | X_i) = G(y_i | x_i, \sigma^2)$
 - $Y_i = X_i + N_i$, where PDF for perturbation N_i is $G(0, \sigma^2)$
 - Perturbation is zero-mean + additive + independent + identically distributed
 - For some noise models (e.g., Poisson, Rician, ultrasound), perturbations aren't independent of signal X_i , or aren't zero-mean
- Let $\theta =$ **parameters** underlying noise model and MRF model (in general)

Image Denoising

- Why do we care about noise model ?

- Row 1: true μ

- Row 2:
 $X \sim \text{Poisson}(\mu)$

- Row 3:
good approx
 - $X \sim \text{Gaussian}$:
Mean = Var = μ

- Row 4:
poor approx.
 - $X \sim \text{Gaussian}$:
Mean = μ ,
Variance = fixed

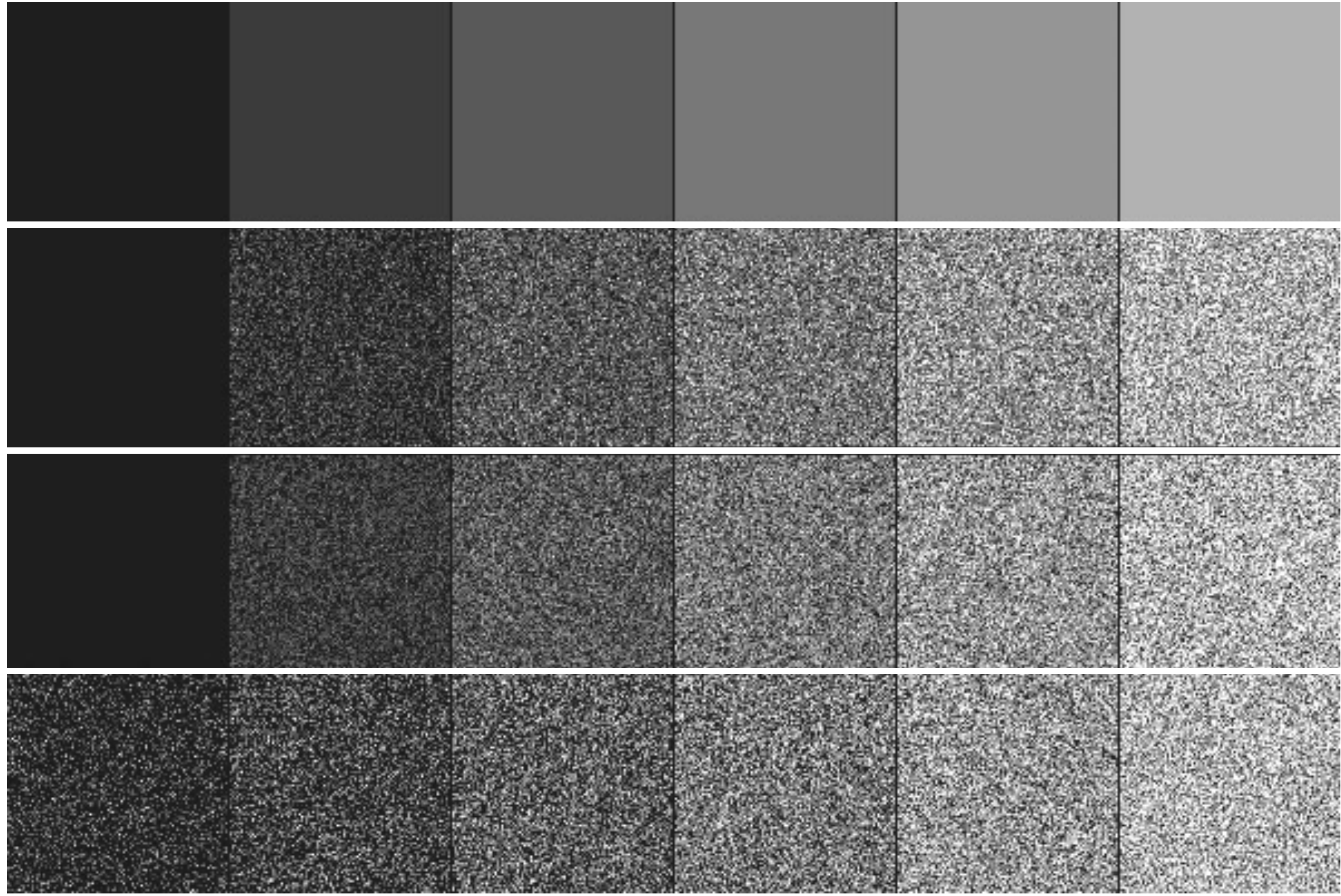


Image Denoising

- Why do we care about noise model ?
- Approximate ways to get to a model of noise variance independent of signal (we don't typically prefer this strategy though)
- Example: “**Variance Stabilization**”
 - Assume variance of X depends on signal value (e.g., Poisson random variable)
 - Find transformation $Y := f(X)$, s.t. **variance(Y) doesn't depend on signal value**
 - e.g., if X is Poisson, then $Z := \sqrt{X}$ makes variance nearly constant
 - Better yet, Anscombe transform $f(X) := 2\sqrt{X + 3/8}$

en.wikipedia.org/wiki/Anscombe_transform

transforms

Poisson random variable X with mean = m

→ approximately-Gaussian random variable with

(a) variance ≈ 1 (when m is large enough) and

(b) mean $\approx 2\sqrt{m + 3/8} - 1/(4\sqrt{m})$

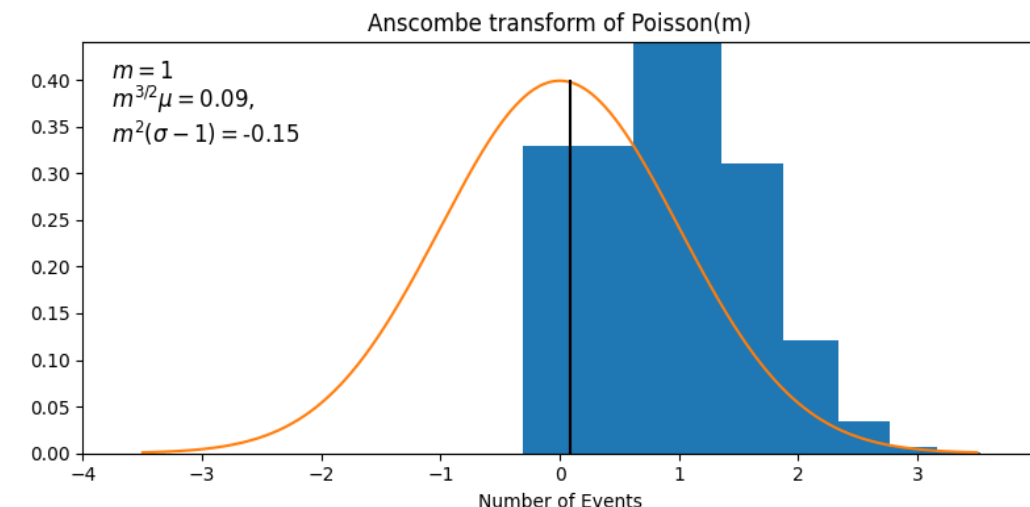
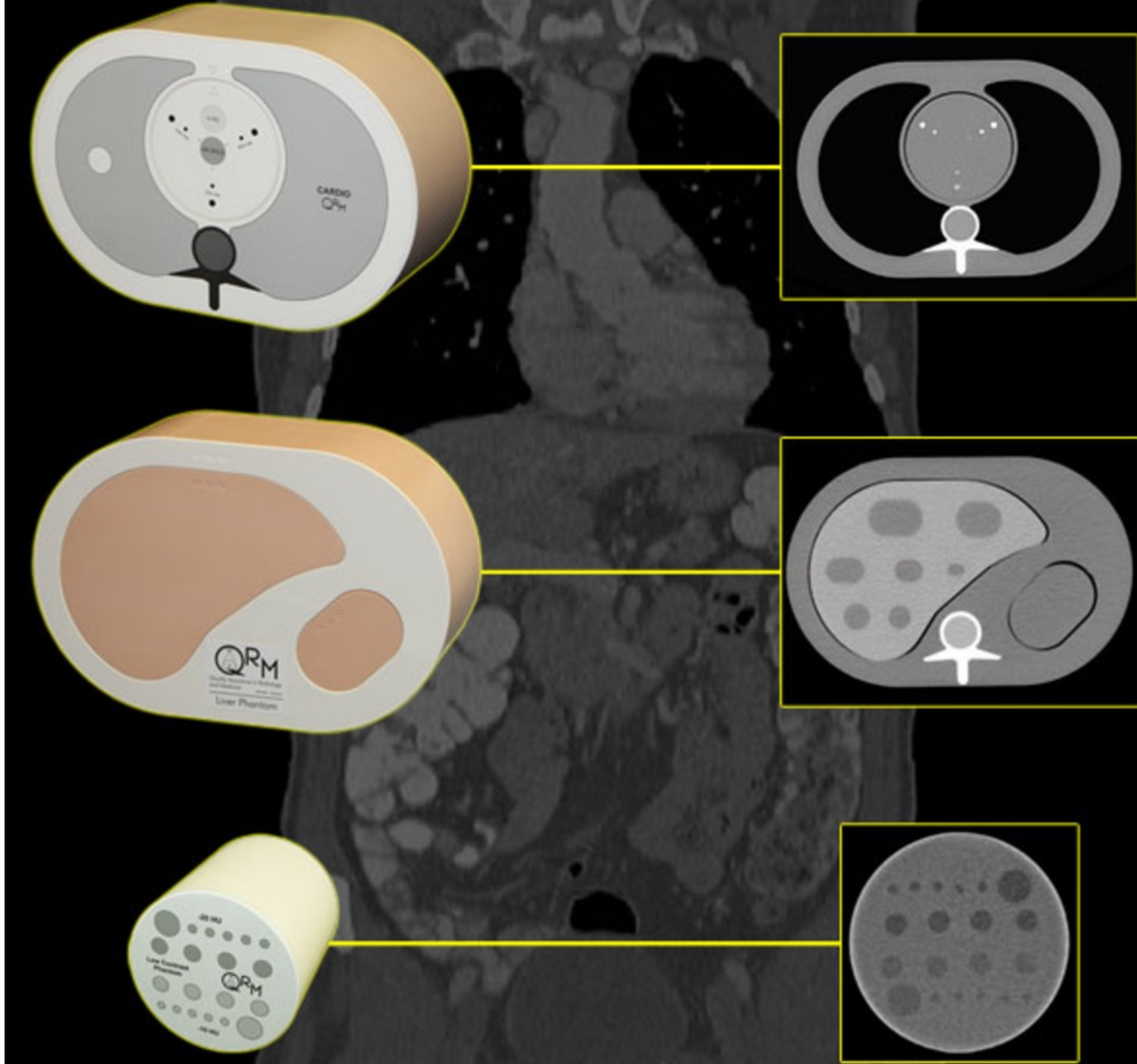


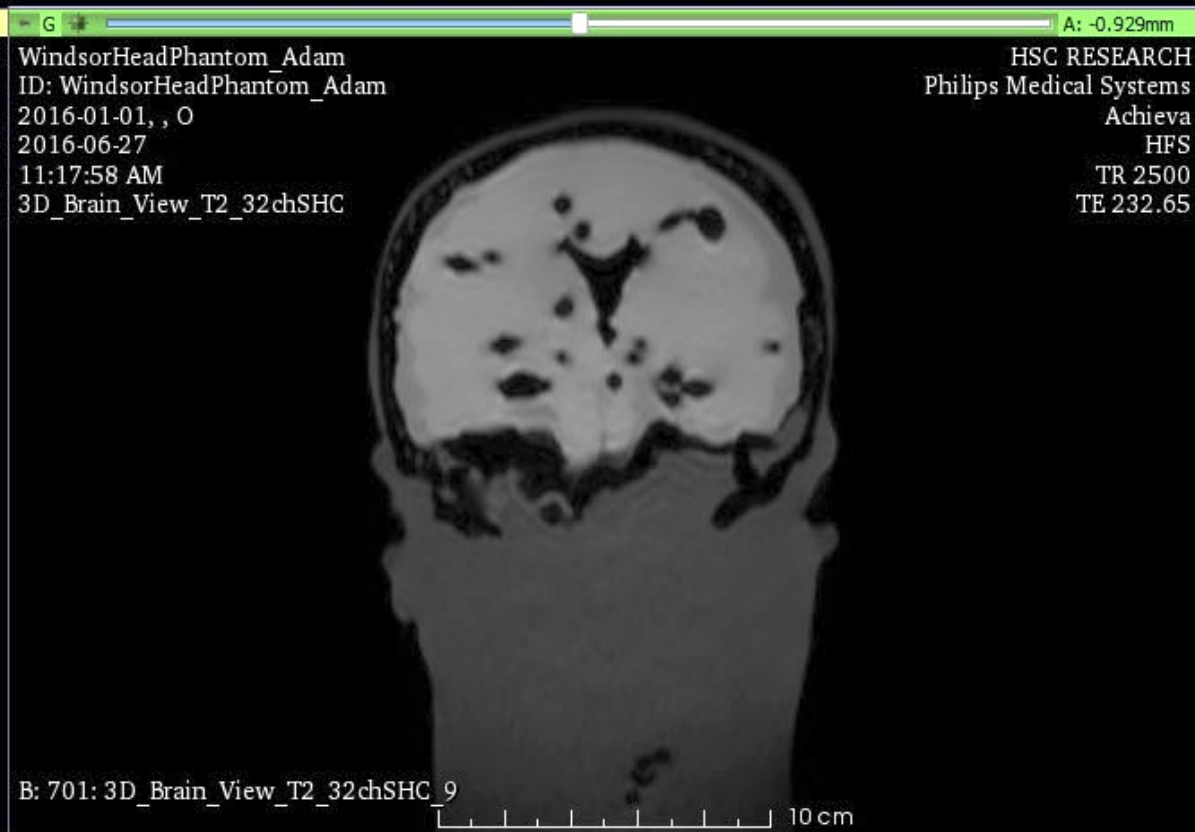
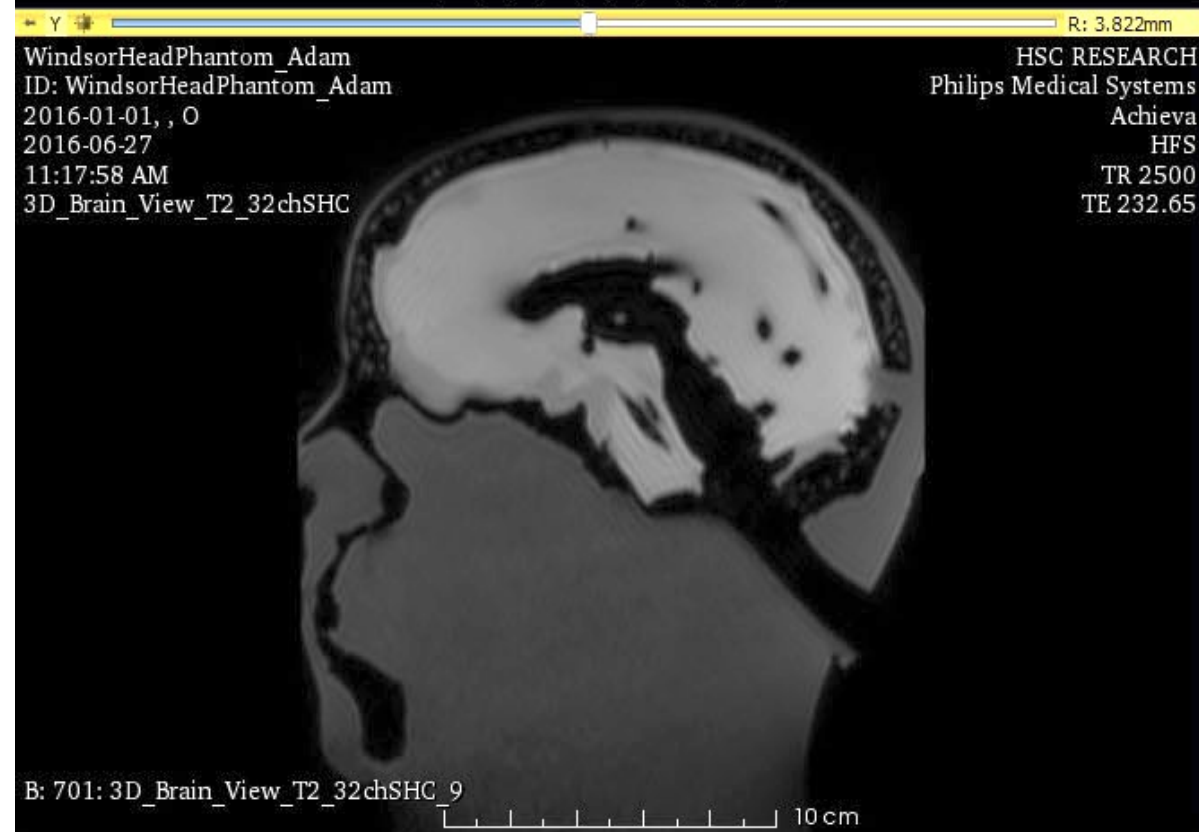
Image Denoising

- Optimization problem and strategy
 - 1) Assume: MRF parameters are user-controlled
 - No need to optimize
 - 2) Assume: noise level is already known
 - Or has been estimated from some **pre-scanned** data
 - a) Scan an “**imaging phantom**”; then analyse image intensities (e.g., perform maximum-likelihood estimation for noise-level parameter)
 - Wikipedia: **Imaging phantom**, or simply **phantom**, is a specially designed object that is scanned or imaged in the field of medical imaging to evaluate, analyze, and tune the performance of various imaging devices.

Imaging Phantom



Imaging Ph



Imaging Phantom



Imaging Phantom



Image Denoising

- Optimization problem and strategy
 - 1) Assume: MRF parameters are user-controlled
 - No need to optimize
 - 2) Assume: noise level is already known
 - Or has been estimated from some pre-scanned data
 - a) Use a prior scan of an “**imaging phantom**”; then analyse image intensities
 - b) Scan “**nothing**” (zero signal); then analyse the image intensities
 - c) Analyse intensities in some image region where signal is known to be zero
e.g., **background** region outside patient body (or **outside tissue** sample)
- Get MAP estimate for noiseless image 'x':
$$\max_x \text{posterior } P(x | y, \theta) = \max_x \text{likelihood } P(y | x, \theta) * \text{prior } P(x, \theta)$$

Image Denoising

- How to optimize ?
- **Simplify** problem: consider task of **only optimizing intensity at pixel i**
 - Assuming all other pixel intensities are fixed (or already optimized)
 - And then **iterate** across all pixels
 - And then make multiple such **passes** over image
- Lets see what happens at pixel i
- Rewrite the objective function
Posterior = $P(X \mid y, \theta)$ as

$$= P(X_i, X_{\sim i} \mid y, \theta)$$

$$= P(X_i \mid X_{\sim i}, y, \theta) P(X_{\sim i} \mid y, \theta) \text{ Conditional Probability}$$

$$= P(X_i \mid X_{N_i}, y, \theta) P(X_{\sim i} \mid y, \theta) \text{ Markov assumption on } X$$

$$= P(X_i \mid X_{N_i}, y_i, \theta) P(X_{\sim i} \mid y, \theta) \text{ Conditional independence assumption in noise model}$$

Image Denoising

- Optimization Algorithm 1
- Consider optimization (mode finding) over a single pixel i
- Performing $\max_{x_i} P(X | y, \theta)$

$$= \max_{x_i} P(X_i | X_{N_i}, y_i, \theta) P(X_{\sim i} | y, \theta)$$

$$= \max_{x_i} P(X_i | X_{N_i}, y_i, \theta) \text{ Second term doesn't depend on } x_i$$

$$= \max_{x_i} \frac{P(y_i | X_i, X_{N_i}, \theta) P(X_i | X_{N_i}, \theta)}{P(y_i | X_{N_i}, \theta)} \text{ Bayes Rule}$$

$$= \max_{x_i} P(y_i | X_i, X_{N_i}, \theta) P(X_i | X_{N_i}, \theta) \text{ Denominator doesn't depend on } x_i$$

$$= \max_{x_i} P(y_i | X_i, \theta) P(X_i | X_{N_i}, \theta) \text{ Conditional independence assumption in noise model}$$

Image Denoising

- Optimization Algorithm 1
- Iterated Conditional Mode (ICM)

1) Iterate over all pixels i :

2) Make multiple passes over image $\max_{x_i} P(y_i | X_i, \theta) P(X_i | X_{N_i}, \theta)$

3) Stopping criterion

- Stop when image hasn't changed much compared to image in previous pass
- e.g., if $\text{RMS}(\text{imageAtPassN} - \text{imageAtPassN-1}) / \text{RMS}(\text{imageAtPassN}) < 0.01$
- 1st term $P(y_i | X_i, \theta)$ = (local) likelihood function $\rightarrow$ Noise model
- 2nd term $P(X_i | X_{N_i}, \theta)$ = (local) conditional (prior) on $X_i \rightarrow$ Image-regularity / smoothness model
- ICM (repeatedly) seeks mode of local conditional (posterior)

Image Denoising

- Various Optimization Algorithms
- We want every update to increase posterior $P(x|y, \theta)$
- We want small running times (should reach optimum in few iterations)
- We don't want optimization algorithm to introduce “artifacts/errors”

1) ICM

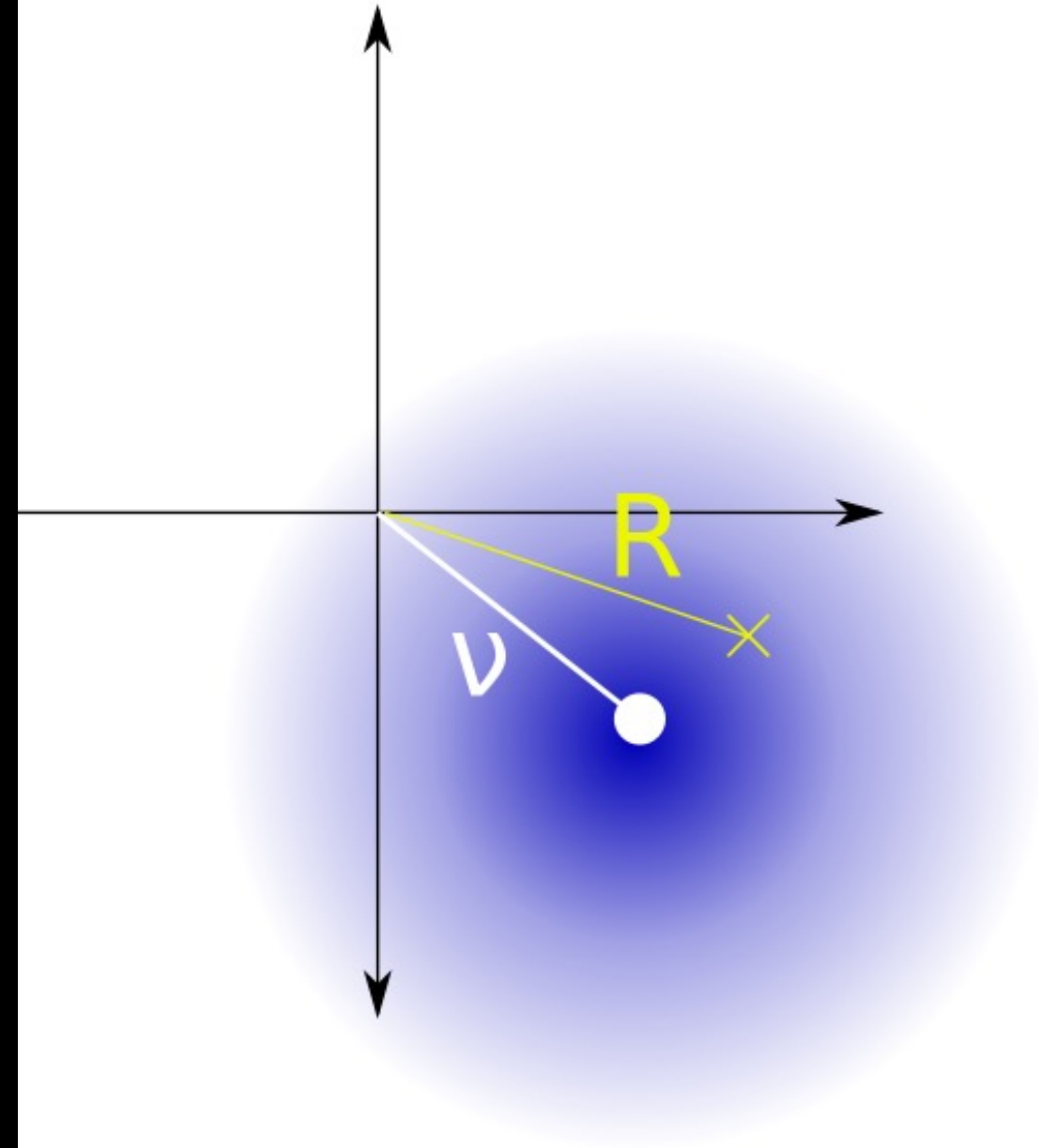
- **Sequentially** visit pixels: **column by column, or row by row**
 - Works in theory, but empirically found to lead to artifacts related to order of visits
- **Sequentially** visit pixels: **randomized order** each iteration
 - Poor locality of memory accesses. Are artifacts eliminated ?

2) ICM-inspired practical strategy

- Step towards local modes (gradient ascent) in **parallel** ; won't guarantee increase in posterior PDF, but take step **if** posterior-PDF **improves**

Image Denoising

- Noise Model: Rician (MRI)



<https://www.vet.upeenn.edu/veterinary-hospitals/ryan-veterinary-hospital/services/radiology-diagnostic-imaging/magnetic-resonance-imaging>

Image Denoising

- Noise Model: Rician (MRI)
- Observed noisy (magnitude-MR) image data y is real (non-negative) valued
- Noiseless (magnitude-MR) image x is real (non-negative) valued
- Prior distribution remains same
- Likelihood model is:

$$P(y|x) := \prod_i P(y_i|x_i) := \prod_i \frac{y_i}{\sigma^2} \exp\left(-\frac{y_i^2 + x_i^2}{2\sigma^2}\right) I_0\left(\frac{y_i x_i}{\sigma^2}\right)$$

where $I_0(z)$ = modified Bessel function of 1st kind, order 0

Image Denoising

- Noise Model: Rician (MRI)

- For ICM optimization, at a chosen voxel i , perform

$$\max_{x_i} P(x|y, \theta) = \min_{x_i} \left(\frac{y_i^2 + x_i^2}{2\sigma^2} - \log I_0 \left(\frac{y_i x_i}{\sigma^2} \right) + \sum_{a \in A_i} V_a(x_a) \right)$$

OR

- For gradient-based optimization with parallel updates, derivative at voxel i is:

$$\frac{\partial P(x|y, \theta)}{\partial x_i} = \frac{x_i}{\sigma^2} - \frac{I_1 \left(\frac{y_i x_i}{\sigma^2} \right)}{I_0 \left(\frac{y_i x_i}{\sigma^2} \right)} \frac{y_i}{\sigma^2} + \frac{\partial}{\partial x_i} \sum_{a \in A_i} V_a(x_a)$$

where $I_1(z)$ = modified Bessel function of 1st kind, order 1

$$P(y|x) := \prod_i P(y_i|x_i) := \prod_i \frac{y_i}{\sigma^2} \exp \left(-\frac{y_i^2 + x_i^2}{2\sigma^2} \right) I_0 \left(\frac{y_i x_i}{\sigma^2} \right)$$

Image Denoising

- Noise Model: Speckle (Ultrasound)



[www.chop.edu/
treatments/
high-resolution-fetal-
ultrasound](http://www.chop.edu/treatments/high-resolution-fetal-ultrasound)

Image Denoising

- Noise Model: Speckle (Ultrasound)

- Observed noisy image data y (real). Noiseless image x (real). Prior is the same.
- Speckle-noise model is: $Y := X + \text{sqrt}(X) Z$, where $P(Z) := \text{Gaussian}(0, \sigma^2)$
- What is $P(Y | X)$? We need this because this is the likelihood

- Use transformation of random variables

$$\text{RV } Z, P(z) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{z^2}{2\sigma^2}\right)$$

$$\text{RV } Y_1 := \sqrt{x}Z, P(y_1) = \frac{1}{\sqrt{x}\sigma\sqrt{2\pi}} \exp\left(-\frac{y_1^2}{2x\sigma^2}\right)$$

Transformation maintains PDF as Gaussian and scales std. dev. by $\sqrt{x}$

$$\text{RV } Y_2 := Y_1 + x, P(y_2) = \frac{1}{\sqrt{x}\sigma\sqrt{2\pi}} \exp\left(-\frac{(y_2 - x)^2}{2x\sigma^2}\right)$$

Transformation maintains PDF as Gaussian and translates mean by x

- So likelihood:

$$P(y|x) := \prod_i P(y_i|x_i) := \prod_i G(y_i|x_i, x_i\sigma^2) = \frac{1}{\sqrt{x}\sigma\sqrt{2\pi}} \exp\left(-\frac{(y_i - x_i)^2}{2x_i\sigma^2}\right)$$

Image Denoising

- Noise Model: Speckle (Ultrasound)
 - For ICM optimization, at a chosen voxel i , perform

$$\max_{x_i} P(x|y, \theta) = \min_{x_i} \left(\frac{\log(x_i)}{2} + \frac{(y_i - x_i)^2}{2x_i\sigma^2} + \sum_{a \in A_i} V_a(x_a) \right)$$

OR

- For gradient-based optimization with parallel updates, derivative at voxel i is:

$$\frac{\partial P(x|y, \theta)}{\partial x_i} = \frac{1}{2x_i} + \frac{1}{2\sigma^2} \left(\frac{x_i^2 - y_i^2}{x_i^2} \right) + \frac{\partial}{\partial x_i} \sum_{a \in A_i} V_a(x_a)$$

$$P(y|x) := \prod_i P(y_i|x_i) := \prod_i G(y_i|x_i, x_i\sigma^2) = \frac{1}{\sqrt{x}\sigma\sqrt{2\pi}} \exp \left(-\frac{(y_i - x_i)^2}{2x_i\sigma^2} \right)$$

Image Denoising

- Motivation for **(re)weighting fidelity** and **regularity** terms
 - 1) Tuning (log)-prior model by **scaling** potential function in GRF
 - 2) Tuning (log)-likelihood model by **scaling** noise level/variance

Image Denoising

- Weighting MRF Prior

- In prior distribution, introduce a parameter $\beta \geq 0$ s.t.

$$P(x) := \frac{1}{Z(\beta)} \exp \left(-\beta \frac{1}{T} U(x) \right) \text{ where}$$

$$U(x) := \sum_{c \in C} V_c(x_c) \text{ where}$$

$$Z(\beta) := \sum_x \exp \left(-\beta \frac{1}{T} U(x) \right)$$

- This changes local conditional (prior) to:

$$P(x_i | x_{N_i}, \theta) = \frac{1}{Z_i(\beta)} \exp \left(-\beta \frac{1}{T} \sum_{a \in A} V_a(x_a) \right)$$

- Interpretation:

introducing β scales potential function $V_a(\cdot)$ to give a new potential function

Image Denoising

- **Weighting Likelihood** (Rician Noise)
- Introduce a parameter $\alpha \geq 0$ where α

$$P(y|x) := \prod_i P(y_i|x_i) := \prod_i \frac{\alpha y_i}{\sigma^2} \exp\left(-\alpha \frac{y_i^2 + x_i^2}{2\sigma^2}\right) I_0\left(\alpha \frac{y_i x_i}{\sigma^2}\right)$$

- Interpretation
 - One way of introducing α is similar to modifying/specifying variance level σ^2 , which keeps us within family of Rician noise models $\rightarrow$ this is Rician PDF with parameters $(x_i, \sigma/\sqrt{\alpha})$
 - On the other hand, if we take $[P(y|x)]^q$, where $0 < q < 1$ or $q > 1$, then resulting PDF doesn't remain Rician

Image Denoising

- **Modified Optimization Problem (Rician Noise)**

- For ICM, at a chosen voxel i , if at all $\beta \in [0,1]$ and $\alpha := 1-\beta$, perform:

$$\max_{x_i} P(x|y, \theta) = \min_{x_i} \left((1 - \beta) \frac{y_i^2 + x_i^2}{2\sigma^2} - \log I_0 \left((1 - \beta) \frac{y_i x_i}{\sigma^2} \right) + \beta \sum_{a \in A_i} V_a(x_a) \right)$$

OR

- For gradient-based optimization with parallel updates, derivative at voxel i is:

$$\frac{\partial P(x|y, \theta)}{\partial x_i} = (1 - \beta) \frac{x_i}{\sigma^2} - \frac{I_1 \left((1 - \beta) \frac{y_i x_i}{\sigma^2} \right)}{I_0 \left((1 - \beta) \frac{y_i x_i}{\sigma^2} \right)} (1 - \beta) \frac{y_i}{\sigma^2} + \beta \frac{\partial}{\partial x_i} \sum_{a \in A_i} V_a(x_a)$$

- $\beta=0$ ($T \rightarrow \infty$) ignores prior; we get ML estimate

- $\alpha=0$ ($\sigma \rightarrow \infty$) makes likelihood uniform

$$\max_{x_i} P(x|y, \theta) = \min_{x_i} \left(\frac{y_i^2 + x_i^2}{2\sigma^2} - \log I_0 \left(\frac{y_i x_i}{\sigma^2} \right) + \sum_{a \in A_i} V_a(x_a) \right)$$

Image Denoising

- **Weighting Likelihood (Speckle Noise)**
- Introduce a parameter $\alpha \geq 0$

$$P(y|x) := \prod_i P(y_i|x_i) := \prod_i G(y_i|x_i, x_i\sigma^2) = \frac{1}{\sigma\sqrt{2\pi}\sqrt{x_i/\alpha}} \exp\left(-\frac{(y_i - x_i)^2}{2(x_i/\alpha)\sigma^2}\right)$$

- Interpretation
 - One way of introducing α is similar to changing noise level / std. dev. σ , which keeps us within family of speckle noise models: this is Gaussian PDF with parameters $(x_i, x_i \sigma^2 / \alpha)$
 - On the other hand, ...

Image Denoising

- **Modified Optimization Problem (Speckle Noise)**

- For ICM, at a chosen voxel i , perform:

$$\max_{x_i} P(x|y, \theta) = \min_{x_i} \left(\frac{1}{2} \log(x_i) + \alpha \frac{1}{2\sigma^2} \frac{(y_i - x_i)^2}{x_i} + \beta \sum_{a \in A_i} V_a(x_a) \right)$$

OR

- For gradient-based optimization with parallel updates, derivative at voxel i is:

$$\frac{\partial P(x|y, \theta)}{\partial x_i} = \frac{1}{2x_i} + \alpha \frac{1}{2\sigma^2} \frac{\partial}{\partial x_i} \frac{(y_i - x_i)^2}{x_i} + \beta \frac{\partial}{\partial x_i} \sum_{a \in A_i} V_a(x_a)$$

- $\beta=0$ ($T \rightarrow \infty$) ignores prior; we get ML estimate

- $\alpha=0$ ($\sigma \rightarrow \infty$) has a non-trivial effect on likelihood

$$\max_{x_i} P(x|y, \theta) = \min_{x_i} \left(\frac{\log(x_i)}{2} + \frac{(y_i - x_i)^2}{2x_i\sigma^2} + \sum_{a \in A_i} V_a(x_a) \right)$$